Data Driven Solutions and Discoveries in Mechanics Using Physics Informed Neural Network

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1. Problem Statement

Many mechanics problems are governed by differential equations. Traditional methods such as finite element and finite difference techniques usually require a mesh and may need a separate method to estimate unknown parameters. In this project, we reproduce three PINN experiments from the reference paper: 1D consolidation, 2D heat conduction, and parameter estimation in a damped dynamic system. We also implement a hard boundary condition for the 2D heat problem, which was suggested in the paper but not experimentally tested.

2. Dataset Details

No external dataset is used. The training points and observations are generated from the governing physical problems.

- Experiment 1: 1D Consolidation: 500 randomly sampled interior points over $x \in (0, 1)$ and $t \in (0, 0.5)$, 250 spatial-boundary points, 125 initial-condition points, and 10,011 test points.
- Experiment 2: 2D Heat Conduction: 1,500 randomly sampled interior points and 500 boundary points over the domain $[-1, 1]^2$, with 10,000 test points.
- Experiment 3: Damped Oscillator: 50 equally spaced, noiseless displacement observations generated from the analytical solution using the true parameters $c = 0.4$, $k = 4$, with $m = u_0 = v_0 = 1$. The experiment also uses 100 random residual points, 50 boundary/initial-condition points, and 10,000 test points. We use $t \in [0, 15]$, inferred from Figure 4 because the paper does not explicitly specify the time interval.

3. Approach

A fully connected network **û(x;θ)** approximates the solution. Derivatives are computed by automatic differentiation, and the governing equation is placed inside the loss: **L = wf ·Lf + wb ·Lb + wd ·Ld**, where Lf is the mean squared PDE residual at collocation points, Lb the boundary/initial-condition mismatch, and Ld the mismatch with observations. In the inverse problem, the **unknown c and k** are trainable variables optimised jointly with the network weights. Training uses Adam, followed by L-BFGS for the two forward problems. Exp1 is non-dimensionalised, and its output is multiplied by x, so **p(0,t) = 0** holds exactly.

4. Implementation Overview

Implemented in PyTorch in a single Jupyter notebook (fixed seed 42, GPU). A shared derivative() helper wraps torch.autograd.grad with create_graph=True for second derivatives. Network sizes, point counts, loss weights, learning rates, and epochs follow the paper: Exp 1 uses 5×50 tanh, Adam 50,000 epochs at 1e-3; Exp 2 uses 4×50, 50,000 epochs at 5e-4; Exp 3 uses 3×32, Adam only, 100,000 epochs at 5e-4. Details the paper omits (L-BFGS settings, test-point layout, split of boundary points across edges, Exp 3 time range, initial guesses for c and k) are our own choices and are marked in the notebook. Extension: Exp 2 with a hard boundary condition, $T = (1-x^2)(1-y^2)\cdot N(x,y)$, trained on the PDE loss alone.

5. Results

Experiment	Metric	Ours	Paper
1. 1D Consolidation	Prediction MSE vs analytical	4.997e-04	not reported
	Max abs. error	3.509e-01	not reported
	Test PDE loss	1.722e+02	not reported
2. 2D Heat (soft BC)	MSE vs FD reference	2.659e-06	not reported
	Max abs. error	1.299e-02	not reported
	Test PDE loss	1.210e-06	not reported
3. Structural Dynamics	c (true 0.4)	0.400816	0.40125215
	k (true 4.0)	3.995299	3.9933197
	Displacement MSE (500 pts)	1.235e-06	not reported
	Test displacement MSE (10,000 pts)	1.256e-06	not reported
	Test ODE loss	2.191e-04	not reported
Ext.: 2D Heat (hard BC)	MSE vs FD reference	4.958e-10	–
	Max abs. error	8.084e-05	–
	Test PDE loss	2.267e-04	–

6. Conclusions

Exp 1: small MSE, but high PDE residual near the sharp $x = 0$, $t = 0$ corner.

Exp 2: closely matched the reference ($MSE = 2.7 \times 10^{-6}$); largest errors occurred near the boundary.

Exp 3: recovered c and k within 0.2% of the true values; convergence depended on initialization.

Hard BC: reduced MSE by about 5000 ×, but increased test PDE residual.

Limitations: single seed, FD instead of FEM reference, noiseless synthetic data.